

Assignment

Phase-2, Week-4, Day-2

Both UI and API

1. Test cases converted and Embeddings in RAG pipeline

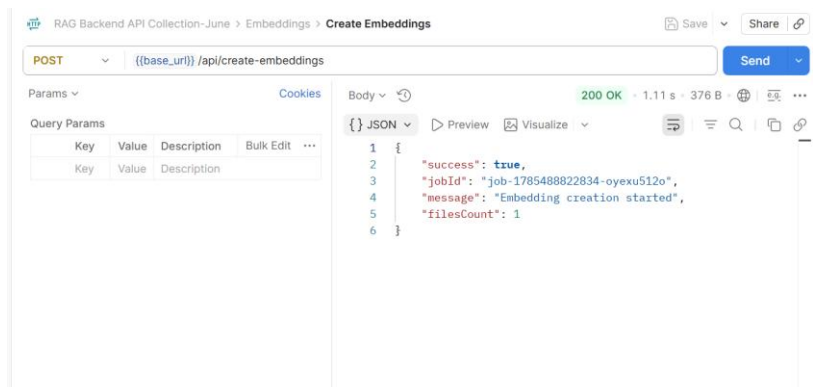
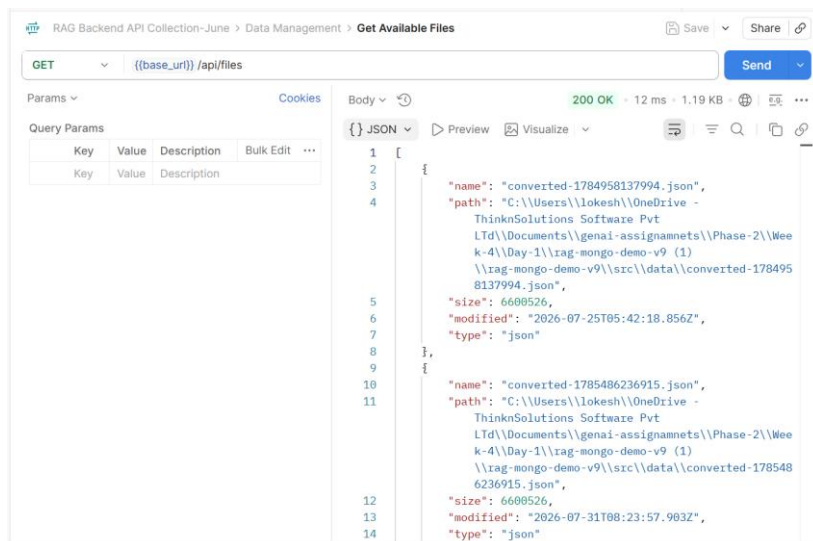
The first screenshot shows the 'Convert to JSON' interface. It includes a navigation sidebar with options like 'Convert to JSON', 'Embeddings & Store', 'Query Preprocessing', 'Vector Search', 'BM25 Search', 'Hybrid Search', and 'Score Fusion'. The main area is titled 'RAG Pipeline' and 'Convert to JSON'. It prompts the user to 'Select data type and configure your Excel file'. The 'Data Type' is set to 'Test Cases'. A file 'testcases.xlsx' (917.3 KB) is selected. The 'Sheet Name' is 'testcases'. A list of required columns is shown: Module, Test ID, Pre-Requisites, Test Title, Test Case Description, Test Steps, Expected Results, Automation/Manual, Priority, Created By, and Created Date. Buttons for 'CONVERT TEST CASES' and 'RESET' are at the bottom.

The second screenshot shows the 'Conversion Results' page. It displays a green success message: 'Success! Test cases file converted successfully'. Below this, the 'Output File' is 'converted-1785486236915.json' (Saved to data folder). The 'Status' is 'Completed' (Ready for embedding creation).

The third screenshot shows the 'Embeddings & Store' page. It displays 'Available Files' with 3 JSON files found. The files are listed in a table:

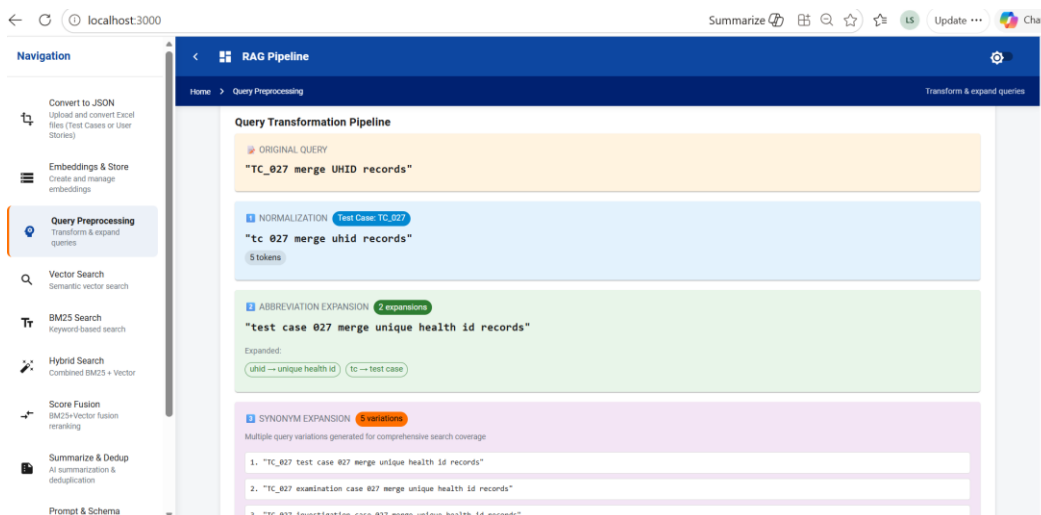
File Name	Size	Modified	Type
converted-1784958137994.jsn	6.29 MB	25/7/2026 11:12:18 am	JSON
converted-1785486236915.jsn	6.29 MB	31/7/2026 1:53:57 pm	JSON
stories-1784958157276.json			

API Get Files and create embedding



2. Query Preprocessing,

- **Normalization** – Convert everything to lower cases
- **ABBREVIATION EXPANSION** – Expands all the abbreviation as per dictionary data setup in code
- **SYNONYM EXPANSION** - Multiple query variations generated for comprehensive search coverage



PIPELINE STEPS	ABBREVIATIONS	SYNONYM VARIATIONS									
Abbreviation Mappings <table> <thead> <tr> <th>Abbreviation</th><th>Full Form</th><th>Position</th></tr> </thead> <tbody> <tr> <td>UHIID</td><td>unique health id</td><td>13</td></tr> <tr> <td>TC</td><td>test case</td><td>-</td></tr> </tbody> </table>			Abbreviation	Full Form	Position	UHIID	unique health id	13	TC	test case	-
Abbreviation	Full Form	Position									
UHIID	unique health id	13									
TC	test case	-									
Before & After Comparison BEFORE (with abbreviations) tc 027 merge uhid records AFTER (expanded) test case 027 merge unique health id records											

API Query Preprocessing

3. Vector search

Sematic search or similar search

The screenshot shows the RAG Pipeline interface with the 'Vector Search' tab selected. The search results are displayed in a table with columns: Test Case, Similarity, Description, and Source. The results are ranked by semantic similarity.

Test Case	Similarity	Description	Source
Transaction Management -> Counter Shift Login	86.5%	Verify user is able to start the shift	OP Billing
Login Details Reports	86.5%	To verify the user able to Delete the role.	OP Billing
UserLogAudit Reports	86.4%	To verify the user able to display Login Success and Fail details	Admin
Registration counter- shift Login	86.3%	Verify user able to assign Duty Roster on Shift Counter Login	Admin

Below the table, there is a 'Detailed View' section showing the 'Transaction Management -> Counter Shift Login' test case details.

API

The screenshot shows the RAG Pipeline interface with the 'API' tab selected. The API endpoint is defined as `POST ((base_url)) /api/search`. The request body is a JSON object containing search parameters like 'query', 'limit', 'filters', 'module', 'priority', 'risk', and 'automationManual'. The response body is a JSON object containing search results, including a list of test cases and their details.

4. BM25

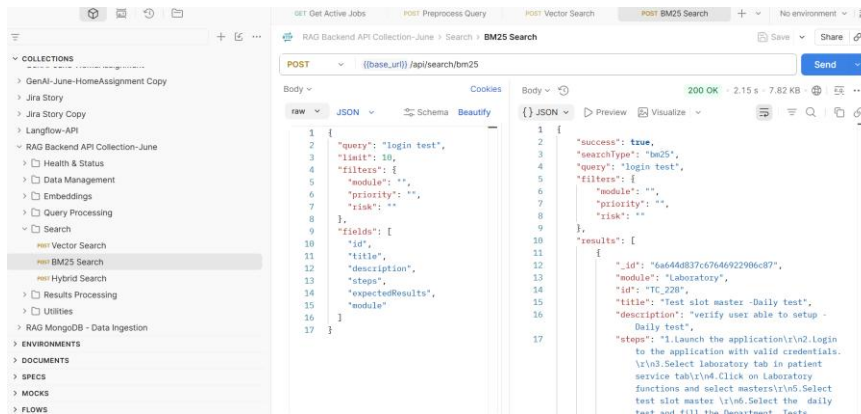
Exact search based on keyword and ranking

The screenshot shows the RAG Pipeline interface with the 'BM25 Search' tab selected. The search results are displayed in a table with columns: Test Case, Similarity, Description, and Source. The results are ranked by keyword-based search using the BM25 algorithm.

Test Case	Similarity	Description	Source
TC_2616	Score: 9.87	Mastersetup->Map Login Counter	Module: AHC

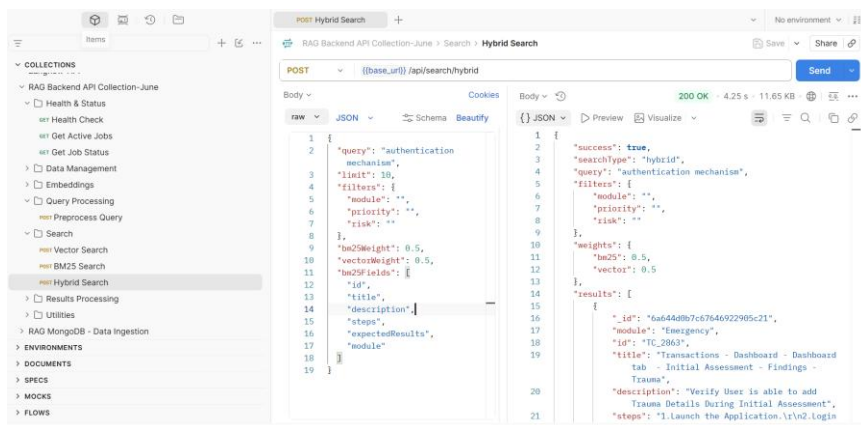
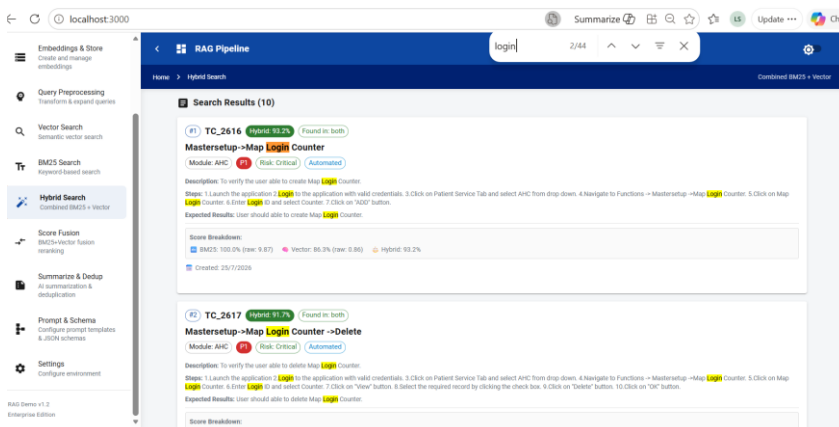
Below the table, there is a 'Detailed View' section showing the 'TC_2616' test case details, including the description and steps to reproduce the issue.

API



5. Hybrid search both vector and BM25

Search based on search weights



6. Score fusion reranking

The screenshot displays the 'RAG Pipeline' interface with the 'Score Fusion' tab selected. The 'Groq AI Reranking Search' section shows a search query 'login details' with a top-k of 50 and a final limit of 10. The results are displayed in a table comparing 'Before vs After Comparison'.

Original Rank (Candidates)	Test Case ID	Title	Vector Score	Groq Rerank Score	Final Rank
#1	TC_3701	Radiology Masters Login Modality Master Save	0.0000	0.0000	#1

7. Deduplication removing duplicate records and summarize using AP

The screenshot displays the 'RAG Pipeline' interface with the 'Summarize & Deduplication' tab selected. It shows a list of test cases with their scores and a section for 'Removed Duplicates (1)'.

Removed Duplicates (1)

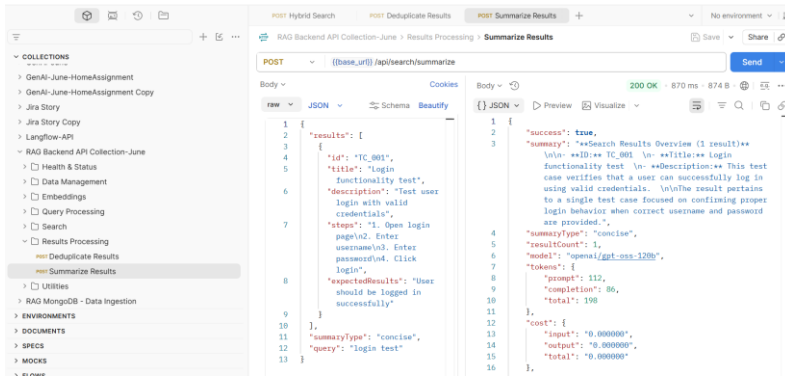
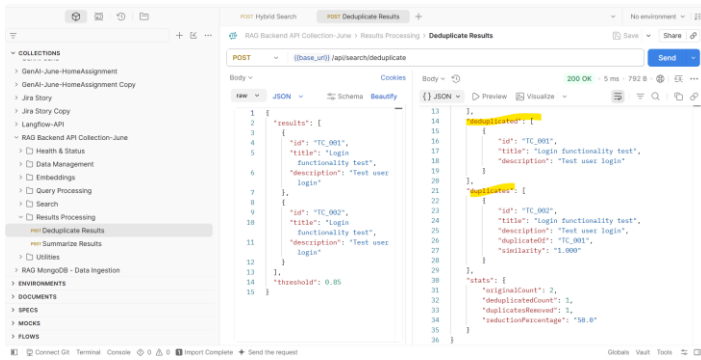
- TC_464 (Duplicate of TC_3646) 100.0% similar
Billing Execution - OP Bill - New Patient-Patient Search

The screenshot displays the 'RAG Pipeline' interface with the 'AI SUMMARY' tab selected. It shows an 'AI-Generated Summary' for 20 test cases.

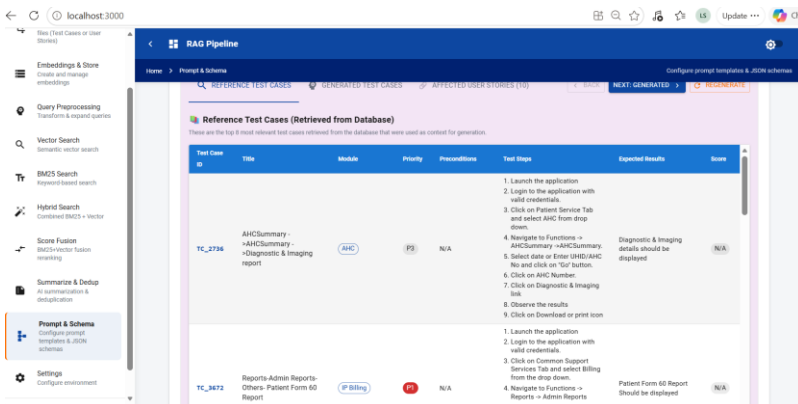
AI-Generated Summary

Search Result Overview (20 Test Cases - ~300 words)

#	Test Case ID	Title (core focus)	Module	Relevance		
1	TC_007	Patient Search	Registration	87.0%		
2	TC_1165	Patient Check-out	IP digital	86.5%		
3	TC_1148	Check-in	IP digital	86.2%		
4	TC_073	Patient Search	Enterprise	Registration	86.2%	
5	TC_023	Patient Program	Registration	86.0%		
6	TC_5064	Doctor Dashboard	In-Patients	Doctors/Wards	86.0%	
7	TC_2616	MasterSetup	Map Login Counter	AHC	86.0%	
8	TC_1144	My Patients List	(nurse allocation)	IP digital	85.9%	
9	TC_4118	Nurse Dashboard	Patient Check-in	Ward Nursing	85.9%	
10	TC_3998	Blood-Bank	Donor Visit	Patient Search	BB (Blood Bank)	85.9%



8. Complete RAG pipeline



SARVAM AI

<https://platform.sarvam.ai/>

1. Chat completion

The screenshot displays the Sarvam AI API interface for a chat completion task. The left sidebar shows a list of collections, with 'Chat Completion' selected. The main panel shows a POST request to the endpoint `/v1/chat/completions`. The request body is a JSON object with the following fields: `model` (sarvam-105b), `messages` (an array of messages), `temperature` (0.5), `max_tokens` (2048), and `stream` (false). The response body is a JSON object with the following fields: `id` (20260731_c5c338b7-eba6-48c0-b1f3-3b45e9fe5f10), `choices` (an array of choices), `finish_reason` (stop), `index` (0), `logprobs` (null), `message` (a message object), `refusal` (null), and `reasoning_content` (a reasoning object). The response status is 200 OK.

2. Text to Audio

The screenshot displays the Sarvam AI API interface for a text-to-audio task. The left sidebar shows a list of collections, with 'Convert (Bulbul v3)' selected. The main panel shows a POST request to the endpoint `/text-to-speech`. The request body is a JSON object with the following fields: `text` (Welcome to Sarvam AI!), `target_language_code` (en-IN), `model` (bulbul-v3), `speaker` (shubh), `pace` (1.0), `speech_sample_rate` (24000), and `audio_format` (wav). The response body is a JSON object with the following fields: `request_id` (20260731_e4533f46-38d1-4929-983d-7da51964bd22), `audios` (an array of audio objects), and `audio_data` (a base64-encoded audio file). The response status is 200 OK.

Base64 to Audio

Convert Base64 to Audio online using a free decoder tool which allows you to decode Base64 as sound file and play it directly in the browser. In addition, you will receive some basic information about this audio file (duration, MIME type, extension, size). And, of course, you will have a special link to download the audio file to your device. If you are looking for the reverse process, check [Audio to Base64](#).

The screenshot shows the Base64 to Audio online decoder tool interface. It features a text input field for the Base64 string, a 'Decode Base64 to Audio' button, and a 'Play Audio' section with a play button and a progress bar. Below the play button, there is a 'File Info' section displaying the duration (1.536s), MIME type (audio/wav), and extension (wav). The interface also includes a 'Comments' section with 63 comments and a rating of 4.3/5.

3. Translate text

The screenshot shows the Sarvam AI API interface for the 'Translate Text' endpoint. The request body is a JSON object with the following fields: 'input' (How are you?), 'source_language_code' (en-IN), 'target_language_code' (hi-IN), 'model' (saivam-translate-v1), and 'mode' (formal). The response body is a JSON object with the following fields: 'request_id' (20260731_80d96d68-fc87-4b40-b44e-017dd1b43a84), 'translated_text' (Aas aap kaise hai), and 'source_language_code' (en-IN).

4. Transliterate

The screenshot shows the Sarvam AI API interface for the 'Transliterate Text' endpoint. The request body is a JSON object with the following fields: 'input' (namaste duniya), 'source_language_code' (en-IN), 'target_language_code' (hi-IN), and 'spoken_form' (false). The response body is a JSON object with the following fields: 'request_id' (20260731_4b7a625e-786e-44c5-b093-8693ed9b744d), 'transliterated_text' (ममस्ते दुनिया), and 'source_language_code' (en-IN).

5. Identify Language

The screenshot shows the Sarvam AI API interface for the 'Identify Language (LID)' endpoint. The request body is a JSON object with the following field: 'input' (यह किस सानसर है:). The response body is a JSON object with the following fields: 'request_id' (20260731_3e54783a-1320-4831-8ae1-f80591ae8abe), 'language_code' (hi-IN), and 'script_code' (Deva).